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Unit Wise Question Bank

Department: Computer Engineering **Semester:I/II** **Academic Year: 2025-2026**

Class:SE **Div: -** **Date:10-9-25**

Course: DELD

Unit No. - name	Q. No.	Question	Marks	CO	Blooms Level
1-	Q.1	State and prove the following Boolean theorems with truth tables: a) A + A' = 1 b) A · 0 = 0	2	CO1	1,3
	Q.2	State and prove the following Boolean theorems with truth tables: a)A(B+C) = AB + AC b)AB+A'C=(A+C)(A'+B)	4	CO1	1,3
	Q.2 a	Explain, in your own words, the logic behind the distributive laws of Boolean algebra by completing and interpreting the results of the truth tables for the following theorems a)A(B+C) = AB + AC b)AB+A'C=(A+C)(A'+B)			
	Q.3	Apply DeMorgan's Theorems to simplify the following Boolean expressions: a) (A + B · C)' b) (A · B · C)'	2	CO1	3
	Q.3 a				
	Q. 4	Convert the decimal number -18 into: a) Signed Magnitude (8-bit) b) 1's Complement (8-bit) c) 2's Complement (8-bit)	4	CO1	3
	Q.5	What are the advantages of 2's complement representation over 1's complement and signed magnitude in digital systems? Explain with an example.	2	CO1	1,4

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	Q. 6	Convert the Boolean expression F = B + A·B' + A'·C into: canonical/ standard SOP form	2	CO1	3
	Q.7	Simplify the SOP expression using Boolean algebra f = A·B + A·B' + A'·B	2	CO1	3
	Q. 8	Convert the Boolean expression F = (A)(B')(A'+B) into: canonical/ standard POS form	2	CO1	3
	Q.9	Convert the Boolean expression F = (A)(B'+D)(A'+B+C') into: canonical/ standard POS form	4	CO1	3
	Q. 10	Simplify the POS expression using Boolean algebra F = (A+B)(A+B')(A'+B)	2	CO1	3
	Q. 11	Simplify the POS expression using Boolean algebra F = (A+B+C')(A+C'+D)(B+C+D)A	4	CO1	3
	Q.12	Convert the Boolean expression f = B + A·B'D + A'·C into: canonical/ standard SOP form	4	CO1	3
	Q.13	Deconstruct or Simplify the SOP expression using Boolean algebra f = A·BCD' + A·B' D' + A'·BC	2	CO1	3
	Q.13 a	Deconstruct the Sum of Product expression in canonical SOP f = A·BCD' + A·B' D' + A'·BC to identify the specific Boolean algebra laws and theorems that can be applied to simplify it. Justify the order in which you will apply these laws, explaining the relationships between the terms and how each step contributes to the overall simplification of the expression.	4	CO1	4
	Q.14	Convert the Boolean expression F = (A+B+C)(B')(A'+C')D into: canonical/ standard POS form	4	CO1	3
	Q.15	Simplify the Boolean function using K-map (4 variables): y = Σ(0, 2, 5, 7, 8, 10, 13, 15)	4	CO1	3

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	Q.16	Minimize the following Boolean function using K-map (3 variables) in POS form: $F(A, B, C) = \Pi(0, 1, 3, 5)$	4	CO1	3
	Q.17	Simplify the Boolean function using K-map $f(A,B,C) = \sum m(1,2,3,6,7)$ Draw the diagram also	6	CO1	3
	Q.18	Minimize the following Boolean function using K-map in POS form: $Y = \Pi M(0, 1, 3, 5,7,9,10,12,14,15)$ Draw the diagram also	6	CO1	3
	Q.19	Simplify the Boolean function using K-map $y = \sum m(1,2,3,6,7). d(0.5)$ Draw the diagram also	6	CO1	3
	Q.20	Minimize the following Boolean function using K-map in POS form: $Y = \Pi(0, 1, 3, 5,7,9,10,12,14,15)+d(2,6,11)$ Draw the diagram also	6	CO1	3
2-	Q.1	Design Half adder	4	CO2	6
	Q.2	Design Full adder	6	CO2	6
	Q.3	Design Half subtractor	4	CO2	6
	Q.4	Design full subtractor	6	CO2	6
	Q.5	Design 8 bit binary adder using 7483	4	CO2	6
	Q.6	Design BCD adder using 7483	6	CO2	6
	Q.7	State rule of BCD addition with examples	4	CO2	1
	Q.8	Why carry lookahead generator is required	2	CO2	4
	Q.9	Write characteristics of excess-3 code.	2	CO2	1
	Q.10	Write applications of grey code	2	CO2	1
	Q.11	Write characteristics of grey code	2	CO2	1
	Q.12	Design BCD to excess-3	6	CO2	6
	Q.13	Design 3 bit binary to grey code	6	CO2	6
	Q.14	Design 3-bit grey code to binary	6	CO2	6
	Q.15	Design excess-3 to grey code	6	CO2	6
	Q.16	Design 2 bit comparator	6	CO2	6
	Q.17	Design 1 bit comparator	4	CO2	6
	Q.18	Implement $f(A,B,C) = \sum m(1,2,3,6,7)$ using 8:1multiplexer	2	CO2	3
	Q.19	Implement 1. $f(A,B,C) = \sum m(1,2,3,6,7)$ using decoder(active low output)	4	CO2	3

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		2. ΠM(0, 1, 3, 5,7,9,10,12,14,15) using decoder			
	Q.20	Implement f(A,B,C) = ∑m(1,2,3,6,7) using 4:1multiplexer	4	CO2	3
	Q.21	Implement f(A,B,C) = ∑m(1,2,3,6,7) using 3:8 decoder (active high output)	2	CO2	3
	Q.22	Design 8:1 multiplexer using 4:1 multiplexer	4	CO2	6
	Q.23	Implement Y = ΠM(0, 1, 3, 5,7,9,10,12,14,15) using 16:1 multiplexer	4	CO2	3
	Q.23	Implement Y = ΠM(0, 1, 3, 5,7,9,10,12,14,15) using decoder (active high output)	4	CO2	3
	Q.24	Implement Y = ΠM(0, 1, 3, 5,7,9,10,12,14,15) using decoder (active low output)	4	CO2	3
	Q.25	Design carry lookahead generator	6	CO2	6

Note: questionstobeaskedinexamarenotlimitedtotheabovesetprovided.

Prof. M.S.Pokale
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